Software Requirements Specification

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines and directs the software development process for the BevOS project in language intended to be comprehensive, clear, and unambiguous regardless of the reader's professional background. This is of particular interest to the target audiences of project leadership (hereafter referred to as the "Project Leads") and the software development team (hereafter referred to as "Development").

1.2 Scope

BevOS is the product to be developed as a software workflow oversight platform that will coordinate information related to the beverage manufacturing process. This first iteration of BevOS will specifically focus on the operations of the company Nice Package Co. a co-packaging company that provides coordination of packaging and manufacturing services to client brands for beverage manufacture and sale. BevOS broadly provides a single webapp for tracking quotes, approvals, and orders into production and logistics. The core services can be broadly broken down into Quotes, Communications (including Approvals Tracking), Production, Logistics, Document Store, and Reports / Analytics as the various domains of information being made available to the users. This information in turn will come from a number of API/data stream points in the domain industry eventually.

1.3 Definitions, Acronyms, Abbreviations

Term	Definition
Approval Workflow	A process used to review and approve operational tasks before they move to the next stage.
BevOS	A centralized workflow-management platform designed for beverage manufacturing operations.
Client	An organization or individual purchasing services or products from another participant in the manufacturing process.
Co-Packer	A third-party company that manufactures and packages beverages for another company.
Core Services	The shared BevOS system software layer that processes requests, applies system rules and coordination functional modules, and manages data exchange.
Dashboard	A visual interface displaying operational information and performance metrics.
Distributor	An organization responsible for delivering products from manufacturers to retailers or customers.
ERP (Enterprise Resource Planning)	Software that integrates core business processes into a unified system.
Logistics	The planning and coordination of transportation and delivery activities.
MFCD (Major Functional Components Diagram)	A diagram illustrating the primary hardware and software components of the system.
Nice Package Company	The beverage-industry client and domain partner for whom BevOS is being developed.
PostgreSQL	An open-source relational database management system used to store application data.
Role-Based Access Control	A security method that grants system permissions according to the user's assigned role and responsibilities.

Single Source of Truth	A trusted, centralized set of current information used consistently by authorized stakeholders.
Stakeholder	A person or organization that participates in or is affected by a beverage-manufacturing workflow.
Workflow	A sequence of business activities completed to achieve a specific objective.

1.4 References

1. Wilson, T. (2026). BevOS Presentation / Nice Package Co. Guest Presentation. CS410 Professional Workforce Development, Old Dominion University.
2. Kennedy, T. (2026). CS410 Recitation Recording and Discussion Materials. Old Dominion University.
3. Grand View Research. (2024, June). Beverage packaging market size, share & trends analysis report, 2030. https://www.grandviewresearch.com/industry-analysis/beverage-packaging-market
4. Honnappa, K. (2026). Platform Visual Aid. Nice Package Co.
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1.5 Overview

The rest of this document details the BevOS system to be developed from a top-down perspective as follows:

Section 2 provides a summary of the background and context of BevOS as a product with a focus on its perspective in comparison to other products, its functionality, the assumed characteristics of its users, and constraints, dependencies, or otherwise imposed from the surrounding environment.

Section 3 provides a list of enumerated requirements organized by the decided feature set of the BevOS system covering the following sub-systems:

2. Overall Description

2.1. Product Perspective

BevOS is an accessory / coordination system purpose built to workflows of the beverage manufacturing industry of Nice Package Co. Other products such as Enterprise Resource Planning (ERP) systems are similar but fall short of addressing the root problems in applying digital solutions to the complex and fragmented processes of the real world commercial and manufacturing domains. BevOS is an exploratory work in software development aiming to balance developer coordination with domain professionals and a design aimed at incremental development as a path towards truly unifying information understanding across the industry.

2.2. Product Functions

BevOS provides a general webapp platform for the coordination of a WAN of devices providing oversight on Quotes, Communications (Approval Tracking, Email), Production, Logistics, Document Store, and Reports / Analytics. This will broadly be covered in Workflow Management through the feature areas of Deals, User Auth and Access Control, Order/Quote Management, Inventory Management, Production Management, Logistics Management, Document Store, and Analytics/Metrics.

2.3. User Characteristics

User Classes can be broadly divided as Brands, Logistics (Co-Packers and Distributors), and Production. Brands mostly are interested in submission and monitoring of requests and market resources while Logistics and Production coordinate more closely with BevOS and by extension one another by also feeding in status information on their own inventory and industrial activities.

In addition to these classes we can identify a class of secondary users in the form of Industrial Workers/Engineers who work with the BevOS system as employees or contractors of the primary users. These users are most concerned with using BevOS to reduce busy work and increase their own labor productivity. They are important voices to guide design and evolution of BevOS as a system.

2.4. Constraints

Due to the Education/Instruction nature of the course this system will aim only to implement a proof-of-concept prototype. While an effort is made to develop novel and creative software solutions, the project's main focus will be on showing the competency of the development team members in showing the professional rigor expected of a Computer Scientist.